A Convolutional Neural Network Architecture for Sentinel-1 and AMSR2 Data Fusion

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***Abstract*— With a growing number of different satellite sensors, data fusion offers great potential in many applications. In this work, a convolutional neural network (CNN) architecture is presented for fusing Sentinel-1 synthetic aperture radar (SAR) imagery and the Advanced Microwave Scanning Radiometer 2 (AMSR2) data. The CNN is applied to the prediction of Arctic sea ice for marine navigation and as input to sea ice forecast models. This generic model is specifically well suited for fusing data sources where the ground resolutions of the sensors differ with orders of magnitude, here 35 km × 62 km (for AMSR2, 6.9 GHz) compared with the 93 m × 87 m (for sentinel-1 IW mode). In this work, two optimization approaches are compared using the categorical cross-entropy error function in the specific application of CNN training on sea ice charts. In the first approach, concentrations are thresholded to be encoded in a standard binary fashion, and in the second approach, concentrations are used as the target probability directly. The second method leads to a significant improvement in R^2 measured on the prediction of ice concentrations evaluated over the test set. The performance improves both in terms of robustness to noise and alignment with mean concentrations from ice analysts in the validation data, and an R^2 value of 0.89 is achieved over the independent test set. It can be concluded that CNNs are suitable for multisensor fusion even with sensors that differ in resolutions by large factors, such as in the case of Sentinel-1 SAR and AMSR2.**

***Index Terms*— Cryosphere, microwave radiometry, synthetic aperture radar (SAR) data.**

I. Introduction

DATA fusion of satellite sensors has a growing potential to improve applications in Earth observation (EO) with the growing number of EO satellites being launched. Observation of the earth with a satellite sensor is often an indirect measurement of the underlying physical/biological process one wishes to estimate, e.g., the satellite sensor measures microwave radiation while we want to estimate ice concentration. The idea of combining measurements from multiple sensors can offer new relationships to the physical process and strengthen the information retrieval.

In this work, a new architecture of a convolutional neural network (CNN) is presented to facilitate the fusion of sensors with a very large ground resolution difference. The architecture differs from other fusion networks in which all sources are not simply resampled and stacked into an extension of image bands before provided as input. Instead, the network saves a large number of computations by concatenating the low-resolution data with the CNN extracted image features at a deeper layer in the network.

While the proposed network architecture is generally useful when sensors have large resolution differences, it is applied here to the problem of generating Arctic sea ice maps from synthetic aperture radar (SAR) images combined with microwave radiometer (MWR) data. In the context of sea ice mapping, SAR images show the spatial patterns formed by sea ice on the ocean in high resolution (<100 m) well, but its backscatter values do not always distinguish between open sea in windy conditions and various ice surfaces. On the other hand, radiometers, such as the Advanced Microwave Scanning Radiometer 2 (AMSR2) that measures brightness temperatures, show a very distinct difference between ocean and ice relatively independent of the wind conditions but at a rather coarse resolution. The potential in combining the information from the two sensors is, therefore, large and will be explored in this work.

SAR satellite images have been used extensively for producing sea ice charts in support for Arctic navigation. However, due to ambiguities in the relationship between SAR backscatter and ice conditions (different ice types and concentrations as well as different wind conditions can have the same backscatter signature), the process of producing ice charts consists of manual interpretation of the satellite data considering also the texture patterns of the ice and ocean in the SAR images. The process is labor-intensive and time-consuming, and thus, the amount of ice charts that can be produced from SAR images on a given day is limited. The long time (often 6–12 h or more) from data acquisition by the satellite to delivery of the ice chart to users also reduces the value of the chart information, especially in dynamic sea ice regions. Automatically generated high-resolution sea ice maps

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can potentially lead to increased use of satellite imagery in ice charting and navigation by producing ice products faster and, more frequently, covering larger geographical areas.

The design of an automatic and robust sea ice classification scheme has been studied for many years [1]. Recent approaches to this issue using CNN (CNNs, as an image segmentation technique) show promising results [2]. In this article, we explore the CNN framework for generating ice products and expand it by introducing a new satellite sensor fusion technique that allows the CNN to exploit MWR data together with SAR images for improved end results.

The contributions of this article are: 1) a general CNN architecture for combining advantages of data acquired using two or more sensors of very different ground resolution; 2) the application of the methodology to automatic generation of sea ice maps; and 3) improvements in CNN training by using "soft" probabilities in applications where these are available, such as the sea ice concentrations (SICs) used here. In terms of the CNN's architecture, the proposed method provides several new concepts. It concatenates very low-resolution sensor data at a deeper layer in the CNN model in order to save convolutions, i.e., computations, rather than stacking all data as an extended band image. With this concept, image features are extracted from the high-resolution sensor and combined with information at a lower scale to improve the pixelwise classification, i.e., semantic segmentation. The concatenation method also enables the inclusion of additional input data that could provide information on the problem at hand, such as wind data in ice charting.

To show the value of the method, it is applied to the case of segmenting sea ice from SAR images. To the authors' knowledge, it is the largest scale experiment in sea ice mapping to date with data from 912 Sentinel-1 scenes distributed over the largest geographical area, covering sea ice conditions in all seasonal variability. It is further suggested to improve the segmentation performance by training the CNN with ice chart concentration as the categorical probability, referred to as "soft" probability, in the loss function rather than binary zero/one encoded target values. The most significant contribution of the study on sea ice mapping is the combination of an SAR system on image level, with radiometer data at metalevel in order to solve the ambiguities in SAR data when distinguishing open sea from ice in certain ice and weather conditions.

II. Data

A data set was prepared by selecting a large set of regional ice charts (also commonly referred to as image analysis, e.g., in [2]) that were manually produced by the ice analysts of the Greenland Ice Service at the Danish Meteorological Institute (DMI) and based primarily on Sentinel-1 SAR imagery. SAR imagery has proven very suitable for Arctic sea ice monitoring due to the radar sensor's capability to see through clouds and in polar darkness. The 912 ice charts and corresponding SAR imagery cover the period from the availability of the Sentinel-1A sensor data in November 2014 and Sentinel-1B data from September 2016, until December 2017, with samples over all months (see Table I). It should be noted that due to the way ice charts that are drawn, they do not necessarily cover a full Sentinel-1 scene. The amount of Sentinel-1 pixels that are covered by an ice chart polygon corresponds to approximately 300 full scenes. Outlines of scenes can be seen plotted on a map in Fig. 1. The individual ice charts cover different areas (depending on users and season), but the data set spans all of Greenland's coast, apart from the northernmost regions, and with denser coverage in Southern Greenland. The combined size of the data set is 156 GB, and it is available for download at [3]. For the studies in this article, data from 88 scenes, approximately 10%, were kept aside from the training process as a test set. It was deliberately chosen to split the data on scene level rather than smaller samples within scene in order to separate training and test sets by either time or geographical region while still making sure that both data sets include all seasons and regions.

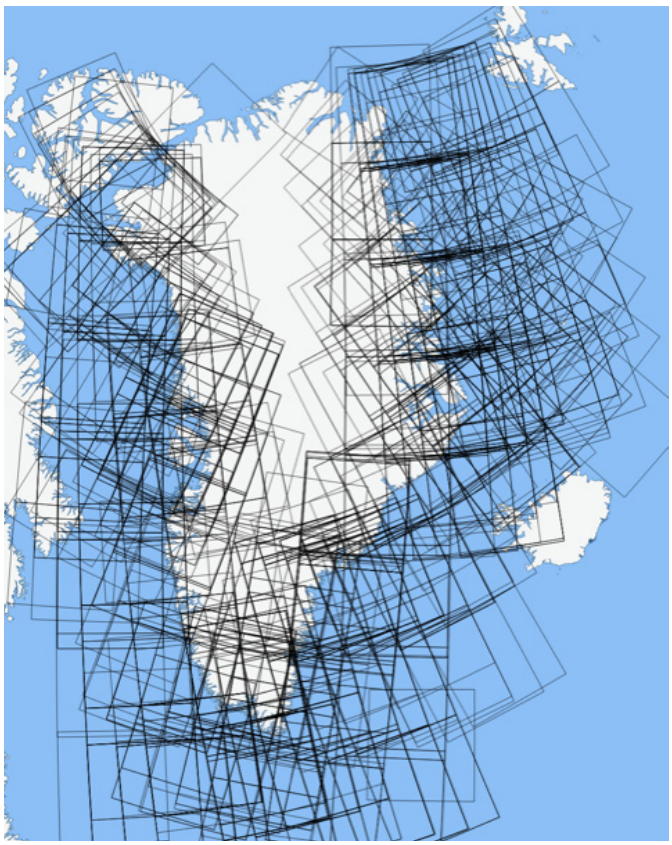

Fig. 1. Outlines of the 912 Sentinel-1 scenes in the ASIP data set covering most of Greenland's coast.

TABLE I
NUMBER OF SCENES PER MONTH IN THE DATA SET

Month	Jan.	Feb.	Mar.	Apr.	May	Jun.	Jul.	Aug.	Sep.	Oct.	Nov.	Dec.
# of scenes	54	38	67	62	74	93	93	115	95	78	70	73

A. Sentinel-1 SAR

We use the C-band SAR data from the Copernicus Sentinel-1 A and B satellites. The data utilized are medium-resolution level 1 ground range detected scenes (GRDM) in extra wide (EW) swath mode with a resolution of

TABLE II
RESOLUTION OF AMSR2 AT DIFFERENT FREQUENCIES [4]. INSTANTANEOUS FIELD OF VIEW (IFOV). P. DIST. IS THE DISTANCE BETWEEN AMSR2 MEASUREMENT POINTS

Freq. [GHz]	6.925	7.3	10.65	18.7	23.8	36.5	89.0
IFOV [km]	35x62	35x62	24x42	14x22	11x19	7x12	3x5
P. dist. [km]	10x10	10x10	10x10	10x10	10x10	10x10	5x5

93 m × 87 m (range by azimuth) and a pixel spacing of 40 m × 40 m. EW GRDM has been chosen because it is the most frequently used mode in marine polar regions with the most favorable combination of coverage and resolution. The Sentinel-1 SAR data are used in their original GRDM geometry in order to avoid resampling the data with potential effects on the image statistics and texture. The data are calibrated to backscatter and converted to decibel (dB) before use. To have a sufficient number of colocated ice charts, we include data from the early period of Sentinel-1 so the HV channel data suffer from quite severe subswath striping and scalloping effects. More recent data (especially after June 2019) include corrections for these effects and will be used in the future when sufficient ice charts become available. The Sentinel-1 SAR data are received in near real time through the Copernicus Marine Environment Monitoring Service (CMEMS) of the European Union. The data are also available from the Open Access Data Hub at https://scihub.copernicus.eu/dhus/#/home.

B. AMSR2

The AMSR2 data are processed by resampling swaths to the coordinates of every 50th pixel in the Sentinel-1 image (corresponding to every 2 km). This results in different oversampling of the AMSR2 frequency bands since each band has a different size footprint (see Table II). In an operational scenario, the Sentinel-1 scene is typically available for the DMI Ice Service 1–2 h after its acquisition as opposed to AMSR2, which is available in near real time. This has been considered when searching for AMSR2 swaths to match with the Sentinel-1 scenes for the data set by searching for the nearest AMSR2 match in time. The AMSR2 data are provided by the Japan Aerospace Exploration Agency (JAXA). We use the L1B data downloaded from ftp://ftp.gportal.jaxa.jp. AMSR2 is an instrument onboard JAXA's GCOM-W satellite (Global Change Observation Mission 1st-Water).

For each Sentinel-1 SAR image in the data set, a matching AMSR2 source is produced by finding the AMSR2 acquisitions which, in the 7-h window preceding the time stamp of the Sentinel-1 pass and 2 h following, intersects with the Sentinel-1 image. Each of the AMSR2 passes in this temporal window and is then resampled onto the subsampled Sentinel-1 pixel coordinates using Gaussian weighted interpolation with the Python *pyresample* library. This results in a time series of resampled AMSR2 data layers, some of which have missing pixel values, due to the Sentinel-1 and AMSR2 swaths not overlapping completely. From this colocated time series, the pixels without missing values and with the smallest time difference to the Sentinel-1 image are selected across the time dimension. In some cases, this results in a mosaic AMSR2 file, where pixels are obtained from two or more swaths. The chosen time window is sufficient to ensure that there are no missing values in this resampled AMSR2 swath, provided that there were no AMSR2 data outages. The median time delay between the nearest AMSR2 and the 900+ Sentinel-1 scenes in the data set is 4 h 45 m, and all scenes are matched within 6-h 40-m time difference.

C. Ice Charts

Manual ice charting from multisensor satellite data analysis has for many years been the common method at the national ice centers around the world for producing sea ice information for marine safety. It should be noted that different ice services have slightly different definitions of the ice chart products and may also refer to them as ice analysis chart or image analysis chart. The ice charts in the data set used here are from the archive of the operational sea ice service at the DMI. DMI has produced SIC charts for the waters around Greenland for more than 50 years. Today, DMI ice analysts primarily use Copernicus Sentinel-1 SAR imagery. The Sentinel-1 imagery used for ice charting is primarily in the EW mode and dual-polarization (HH+HV). Auxiliary satellite data, such as optical imagery, thermal-infrared and passive MWR data, and numerical weather prediction data, are used when available to support the analysis. The ice chart coverage and frequency depends on season, ship routes, and data availability. Normally, one ice chart is produced on the basis of one Sentinel-1 image and is thus a snapshot of the ice conditions at the time of the SAR image acquisition. The ice analysts use the latest available Sentinel-1 SAR image and carry out a detailed manual interpretation by drawing the ice chart in an ArcGIS production system. An ice chart consists of manually drawn polygons of fairly homogeneous ice conditions given as a concentration of ice inside the polygon from 0% to 100%, where 100% ice is completely covered ocean and 0% ice means open water. The estimates of ice concentration in the charts are based on the subjective evaluation by the analyst and individual ice charts or polygons have no associated uncertainty. Analysts pay particular attention to regions near the ice edge because the characteristics and extent of ice in the marginal ice zone are important for operations taking place within or near that region. Conversely, the DMI analysts generally do not characterize the inner ice zone with as much attention to detail because most of the time there are no supported navigation there. It is important to notice that the relative accuracy and level of analysis detail varies. Studies of ice charts from different ice centers covering the same geographical region show relatively large (up to 20%) variability in ice concentrations [5]. The original ice charts can be found in the ice chart archive at the http://ocean.dmi.dk/arctic/icecharts_gl_centralwest.uk.phpDMI homepage. The ice charts follow the World Meteorological Organization's (WMO) code for sea ice, where the ice concentration classes are defined by concentration intervals, spanning over 10%–30%. When converting the original ice chart format into training data for this study, these concentration intervals are converted to their mean ice

concentration. The range between the concentration values is, therefore, 5% (e.g., 0%, 5%, 10%, 15%,…, up to 100%). The ice concentrations are seldom homogeneously distributed inside the drawn polygons, meaning that the concentration of a smaller subset of the polygon, e.g., a 40 m × 40 m pixel, does not have the same concentration as the polygon it lies within. The average concentrations of subsets are considered correct for the polygon, but the distribution of ice that averages to the given percentage can vary a lot. Ice charts are produced for areas where there is maritime traffic, and since the majority of the ships in Greenland waters try to avoid sailing in sea ice, the ice maps are focused on showing areas of open water in between sea ice. In eastern and southern Greenland sea ice, drift along the coast and the ice charts covering these regions often has large open water polygons and a narrower area of sea ice. These ice charting practices lead to the data set having an overrepresentation of open water samples of ∼70% and the remaining ∼30% being ice samples distributed over other concentration values, >0%.

Due to the relatively poor quality of the Greenland coastline data used in the ice charting software (with offset and missing islands), a land-mask buffer zone of 10 km is applied to the ice chart 2-D variable in the training and test data. However, the final CNN model is capable of producing results much closer to the coast.

III. Related Work

A. Satellite Sensor Fusion Techniques

The overall objective of satellite sensor data fusion is combining advantages from two or more sensors for improving derived data products compared with using a single sensor only. With a vast amount of EO sensors operating today, the potential for development in sensor fusion techniques is larger than ever. The fusion of data from satellite sensors has been used to address a wide range of problems through time, and numerous techniques exist to accomplish this. For an overview of the field, we refer to the following review papers [6]–[8]. In satellite sensor fusion, one distinguishes between three overall types, pixel level, feature level, and decision level [6].

In pixel-level fusion, two or more coregistered sources, typically images, are combined to reveal more details than what was available from a single source. An example is pan-sharpening of multispectral images where the goal is to achieve as high spatial resolution as the pan-chromatic band for all multispectral bands in the sensor, e.g., as in [9].

Feature-level fusion is the concept of first extracting features from different sources and then using them as inputs to statistical learning models. One example is the land cover classification where fusion of SAR and optical images can lead to improved results [10].

In decision-level fusion, each source of data has undergone processing and preliminary analysis, such as classification. An example of multisensor fusion at the decision level is shown in [11] where several preclassified data sources are combined and compared with different weighting schemes.

Some of the challenges in multisensor fusion are the spatial coalignment of the data sources as well as the potential temporal gap between the recordings. Another challenge is how to fuse the data from sensors with a very large difference in spatial resolution. We address this last challenge by utilizing the CNN model's architecture that handles this in a computationally elegant way. We propose a CNN architecture capable of handling data from different sources where the spatial sampling resolution is many factors higher in one sensor. For the case of the Sentinel-1 and AMSR2 fusion, the resolutions differ with magnitudes.

The proposed model in this article takes Sentinel-1 and AMSR2 as input. Our model does a number of filter/convolution operations on the SAR image before injecting the MWR data, and the approach can, therefore, be seen as a feature-level fusion strategy. The proposed model is described further in Section IV and shown in Fig. 2.

B. CNN Segmentation

With the increased use of deep learning in a wide range of areas in recent years, one of the promising applications of CNNs has been obtaining improvement for image segmentation (here defined as splitting an image into "segment" by pixelwise classification) [12]. The convolutional layers of a CNN that applies a number of filters to an input image (or other data types) are well suited to learn how to extract useful features for classifying information, either at image level or pixel level. The filters in a network are ordered hierarchically by stacked layers each consisting of a filter bank. For classification, information from all pixels is reduced to a single label, and layers are therefore interleaved with subsampling schemes that reduce the spatial extent and make the model invariant to translations of the objects in the image. In the context of CNN image segmentation, the use of subsampling comes with a tradeoff on the models' ability to create sharp edges around objects in the output segmentation map. Currently, two main concepts of CNN architecture are used for image segmentation across many fields of use. The first, named *U-Net*, is based on the work in [13], which was further refined with "skip connections" in [14]. In the *U-Net* architecture, convolutional layers and subsampling schemes are applied to an input image similar to traditional CNN architectures for classification tasks. The subsampling reduces the spatial dimension and allows for more convolutional filters to be trained using the same or fewer computations. Another advantage of subsampling in terms of segmentation is the relative increase of the "receptive field," i.e., is the amount of neighborhood pixels in the input image considered in the classification of each pixel in the output. When applying subsampling, such as a 2 × 2 max-pooling function, the following convolutional filters consider a factor of four times larger area from the input image, i.e., the receptive field grows by a factor of 4. The size of the "receptive field" is considered important to capture the spatial semantics in the segmentation tasks. In order to go from a low spatial dimension back to a segmentation map with the same spatial dimension as the input image, *U-Net* applies upsampling operations interleaved with convolutional layers. However, subsampling corresponds to low-pass filtering, and

thus, high-frequency image details are lost in this process. For retaining sharper edges, i.e., high-frequency information in the final output segmentation map, "skip connections" before each subsampling step is concatenated to the corresponding upsampling step that has the same spatial dimension. *U-Net* was originally developed for a medical image application, but it has now proved useful in a far range of fields and applications.

As opposed to the U-Net, the second concept of CNN segmentation architectures, introduced in [15] and [16], does not apply any subsampling operations. This is to avoid the low-pass filtering property, but the high spatial dimension throughout the layers leads to a relatively higher computational cost on average per layer. To retain a "receptive field" similar to that of a CNN that includes subsampling, *Atrous* convolutions are applied. In an *Atrous* convolution, the kernel coefficients are separated with a certain number of zeros or holes (from French *à trous*: with holes) in order to have a computational low-cost kernel that extends over a larger area in the input. The *Atrous* convolution thereby increases the receptive field and eliminates the need for spatial subsampling for incorporating image features at larger scales. *Atrous* filters were originally introduced by Holschneider *et al.* [17] for scale-dependent wavelet transforms.

Chen *et al.* [15] evaluated four different strategies for incorporating scale space in a CNN model and concluded that a strategy they named Atrous spatial pyramid pooling (ASPP) performs best. An ASPP CNN model applies first regular convolutional layers and then four parallel *Atrous* convolutions with different dilation rates, i.e., different spacing between coefficients. The output of the four *Atrous* layers is concatenated, and a 1×1 convolution is applied to produce the output segmentation map. The model achieves an image feature extraction at different scales and is known to be improving context modeling in image analysis [18].

Except for obvious numerical and computational differences regarding the use of subsampling in a CNN or not, the largest conceptual difference between the two concepts lies in how they handle scale-dependent information. In a U-Net style network, features from different scales are added from coarse scales to finer grained information in the stepwise upsampling process of the so-called decoder part of the network. In the *Atrous* style network, scale information can be handled either by applying the *Atrous* operator throughout a network on sequentially stacked convolutional layers or alternatively by an ASPP concatenation module. The latter is chosen in this work due to the concatenation layer allowing for a direct extension beneficial for our fusion methodology, which is further explained in Section IV.

C. Automating Sea Ice Products

Previous literature on automatic extraction of sea ice information from satellite data takes different approaches to obtain training and validation data. The most common automatic product is derived from MWR data alone, and very high accuracy can be achieved [19]. However, the coarse resolution of the MWR SICs is a major drawback when it comes to usage for navigation where more details are desired. Until recently, attempts to automatically derive SIC from SAR data have been using precompiled texture measures such as based on gray-level co-occurrence matrix methods [2]. Often, ice charts produced by expert ice analysts are the only source of the target information, and although as described in Section II-C, these products contain uncertainties in terms of interoperator/intraoperator variability, they serve as the only way of data annotation. Furthermore, in [20], it is described how some ice services deviate from the WMO's guidelines. As opposed to the WMO guidelines, the approach of the Finnish Meteorological Institute described in [20] attempts to increase the consistency of the labeling process of polygons in ice charts by always assigning the same information on ice. Wang *et al.* [2] turned the problem into a CNN least-squares regression task over SICs with RADARSAT-2 images as input. The authors work on patches of 45×45 pixels that have a pixel spacing of 400 m giving a patch size of 18 km. When evaluating new scenes, the network is applied on a sliding window manner over all pixel positions to yield a higher output resolution and a smooth looking prediction. When doing regression over SICs, the final resolution cannot be too fine as it will introduce a large amount of representation errors. This is due to the fact that ice charts consist of large polygons, each labeled with its average ice concentration only. If one looks at a small sample from a polygon, the true ice concentration can differ quite a lot from the polygon average SIC value compared with the ice chart.

Wang *et al.* [2] and Karvonen [20] used two different modalities of neural networks, CNN, and multilayer perceptron, respectively. They both use the least square error loss function to optimize network parameters, which corresponds to a maximum likelihood optimization of NNs given a normal distributed prediction error [21]. Whether the Gaussian assumption is valid for SICs is not reported, but both papers reach convincing results in terms of overall SIC maps. Karvonen [20] stated that SIC levels below 25% have high uncertainty and found that SICs below 25% often are open water. Therefore, [20] suggested thresholding of SICs $< 25\%$ and set everything below to 0.

As an alternative to a regression task, one could consider predicting an ice mask with a classification loss function and maximize categorical probabilities of ice versus water. This approach is adopted by Zakhvatkina *et al.* [22], Leigh *et al.* [23], and Wang *et al.* [24]. Zakhvatkina *et al.* [22], Leigh *et al.* [23], and Wang *et al.* [24] developed algorithms that classify ice/water based on support vector machines. The final product resolution in [22]–[24] range between 200 m and 16 km.

IV. CNN Architecture

The design goal of the proposed CNN architecture is to incorporate certain capabilities. The first, is the fusion of two satellite data sources, the Sentinel-1 SAR data and AMSR2 MWR measurements, both measuring distinctively different electromagnetic properties on earth at very different spatial resolutions. The second is to capture spatial patterns, i.e., image features, in the SAR image at different scales

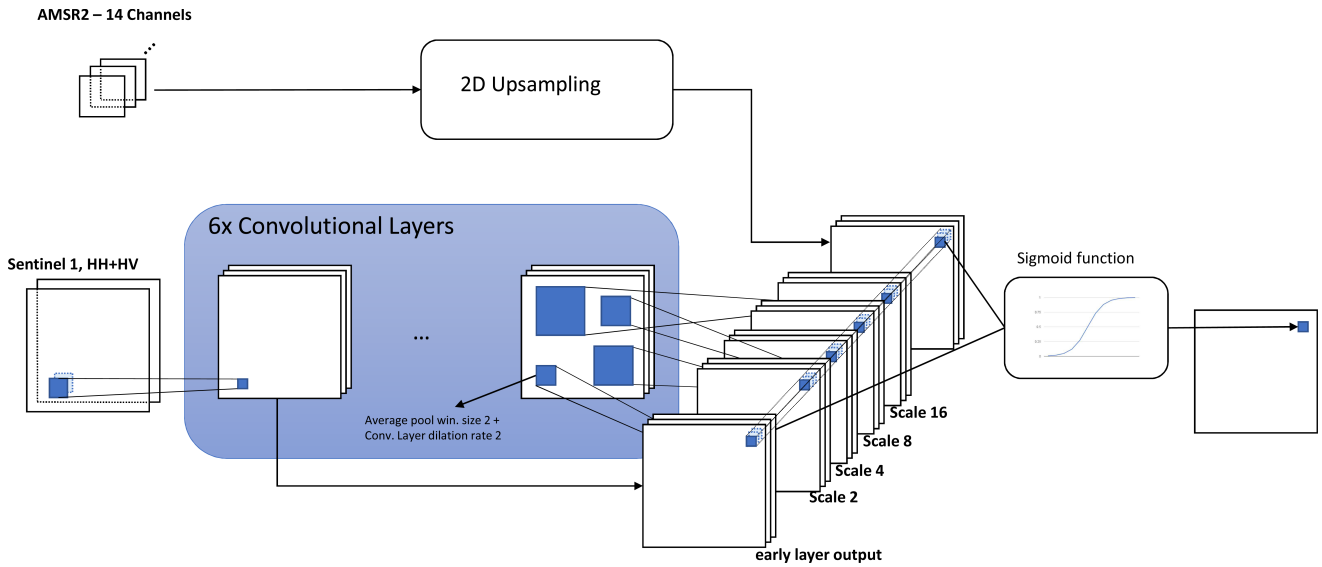

Fig. 2. CNN architecture. The 2-D upsampling performed on the AMSR2 data inside the CNN is done in order to implement the last layer as 1×1 convolutional filter kernels.

in order to retain very high resolution, i.e., sharp edges, as well as mapping the large-scale features that distinguish ice from open water in windy conditions. The third is to retain the absolute backscatter value in the SAR images while letting the CNN extract optimal feature descriptors. When using batch normalization layers in the CNN, where a standard normalization is applied between layers, the scaling is performed on a small batch of data and the relative backscatter values between batches might be shifted. Batch normalization is a regularization technique that is known to accelerate the training and allow for deeper networks since gradients are controlled by scaling [25]. By letting the output of the second convolutional layer, which has not undergone batch normalization yet, circumvent the rest of the layers and adding it to the concatenation just before the output layer, we ensure that these backscatter values have not been scaled in a batchwise manner. If any information on the sea ice problem lies in the absolute value of the SAR signal, we ensure that this is kept in the classifying output layer. According to [26], the simplest way of performing data fusion with CNNs is by stacking the different sources and thereby extending, e.g., the RGB image to more channels, as done in [27] with RGB data and a digital elevation model. In the case of very different spatial resolution as with Sentinel-1 SAR and AMSR2 microwave radiometry, stacking the sources leads to an enormous amount of unnecessary redundant convolutions performed on many channels, as data must be resampled to the highest resolution before they are fed to the network. The architecture used in this work, therefore, injects the low-resolution AMSR2 data at a later stage in the CNN, as shown in Fig. 2, and in this way, it fuses the sources intelligently. Sentinel-1 SAR input data are therefore handled separately with six pairwise 3×3 kernel convolutional layers with batch normalization and dropout in between. To model the image data at different scales, the ASPP module is adopted in the second to last layer of the model. The ASPP module extracts the features at four different scales by performing *Atrous* convolutions with four different dilation rates and concatenating the outputs. Thereby, features are extracted at both coarse and finer levels and it is up to the last layer to mix these in a way that is optimal for the optimization problem. Average pooling layers, i.e., 2-D moving average filters, were applied with window size equal to each of the four dilation rates. This was necessary to avoid artifacts in the predictions due to the "hole" between filter coefficients in an *Atrous* filter.

The input size of the Sentinel-1 image patch, as well as the output prediction mask, is 300×300 pixels, i.e., 12 km $\times$ 12 km, of each HH and HV channel. The corresponding ASMR2 resampled values are 6×6 pixels of 14 channels consisting of seven frequencies in two polarizations brightness temperatures. After training, the CNN on 282 948 training samples cuts out without overlap from 824 SAR scenes in the training set, the model can be applied on a new unseen SAR/AMSR2 scene in sliding window manner. Each window is overlapped with 50 pixels, and the 50-pixel edge around each prediction output is discarded to avoid predictions with padded values. We trained all models with the Adam optimizer and left the values of its parameters as advised in [28]. Training was performed for a fixed 80 epochs for all models as this generally ensured a good convergence was reached. Code for the model definition can be found at [29].

A. Optimization Strategies

The choice of final product resolution and whether to model the problem as regression over SIC or classification of ice/water is connected to the end-user need. Typically, SICs are provided over large polygons that will indicate zones where ships of certain ice grade can or cannot navigate, but these products are not always sufficient. In terms of ice/water products, these would often be needed in much higher resolution as

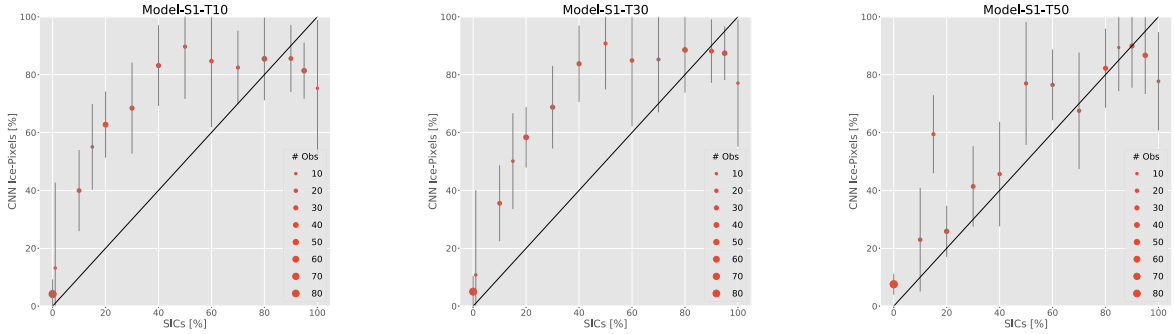

Fig. 3. Summary of testing against expert labeled ice concentrations when the CNN model is trained on, i.e., labels from the ice charts being thresholded at a given concentration level. Y-axis is the CNN predictions, fraction of pixels classified as ice within each polygon, and X-axis is the SIC value from the ice chart. Ice concentration levels with less than five examples in the test set have been excluded. The red dots are mean over all polygons in the test set and the gray line is the standard error. (a) $T = 10\%$. $R^2 = 0.33$. (b) $T = 30\%$. $R^2 = 0.39$. (c) $T = 50\%$. $R^2 = 0.74$.

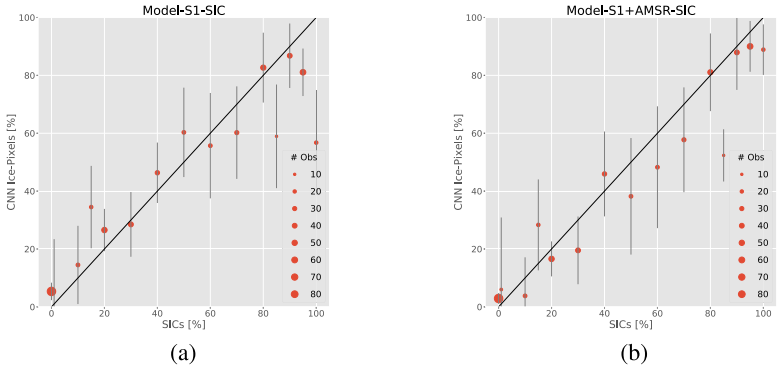

Fig. 4. Summary of testing against expert labeled ice concentrations when the CNN model is trained on categorical probabilities being set to the ice concentration level from the ice charts. Y-axis is the CNN predictions, fraction of pixels classified as ice within each polygon, and X-axis is the SIC value from the ice chart. The red dots are mean over all polygons in the test set and the gray line is the standard error. (a) Without AMSR2 data ingestion. $R^2 = 0.8$. (b) With AMSR2 data ingested. $R^2 = 0.89$.

it might be used, e.g., by fishing vessels that want to navigate as close to the ice edge as possible without encountering dense ice. As our user survey (report available in Danish at [30]) showed a demand for very high-resolution products, our method is based on a CNN model where presubsampling is not necessary. This allows for experiments with final product resolution as high as possible and for the model to be transferred to even higher resolution sensor data if this would become available in the future. In terms of training a CNN to classify pixels, the categorical cross-entropy loss function for maximum likelihood classification is used as written as follows:

$$E(\mathbf{w}) = \sum_i t_i \log(y(\mathbf{w}, x_i)). \tag{1}$$

Here, $y(\mathbf{w}, x_i)$ is the output of the CNN for the i'th sample given the weights $\mathbf{w}$ and t_i is the target for the i'th sample. The most common way to encode t_i is [0; 1], where 1 indicates ice and 0 open water. For the ice charts to provide this encoding, they must first be transformed to zeros and ones by applying a threshold at a certain SIC value, as in [22] and [23]. This will introduce some amount of label errors as a pixel within a polygon in the chart might be open water even though the mean ice concentration is high or vice versa.

TABLE III

PIXELWISE ACCURACY OF THE FOUR MODELS WITH THE ICE CHART BEING THRESHOLDED AT 10% CONCENTRATIONS TO CONVERT THEM TO BINARY LABELS FOR ICE/WATER. EVALUATED WITH THE TEST SET OF 28 651 IMAGE-WINDOWS OF 300 × 300 PIXELS. T_p IS THE THRESHOLD ON THE CNN OUTPUT PROBABILITY AT WHICH WE CONSIDER A PREDICTION ICE

Name	Model-S1-T10	Model-S1+AMSR-T10
Input	S1	S1 + AMSR2
Opt. Method	ice charts thresholded at 10%	
Acc., $T_p = 10\%$	86.82%	89.85%
Acc., $T_p = 30\%$	91.24%	94.76%
Acc., $T_p = 50\%$	**91.81%**	**95.29%**
Acc., $T_p = 70\%$	91.47%	94.91%
Acc., $T_p = 90\%$	89.07%	92.93%

Name	Model-S1-SIC	Model-S1+AMSR-SIC
Input	S1	S1 + AMSR2
Opt. Method	SIC used as categorical probability	
Acc., $T_p = 10\%$	**90.64%**	**94.26%**
Acc., $T_p = 30\%$	88.5%	89.46%
Acc., $T_p = 50\%$	81.39%	82.40%
Acc., $T_p = 70\%$	73.51%	75.85%
Acc., $T_p = 90\%$	55.89%	64.27%

Neural networks are known to cope well with some amount of label errors due to the stochastic nature of the training process [31]. Alternatively, we propose to use the SICs as

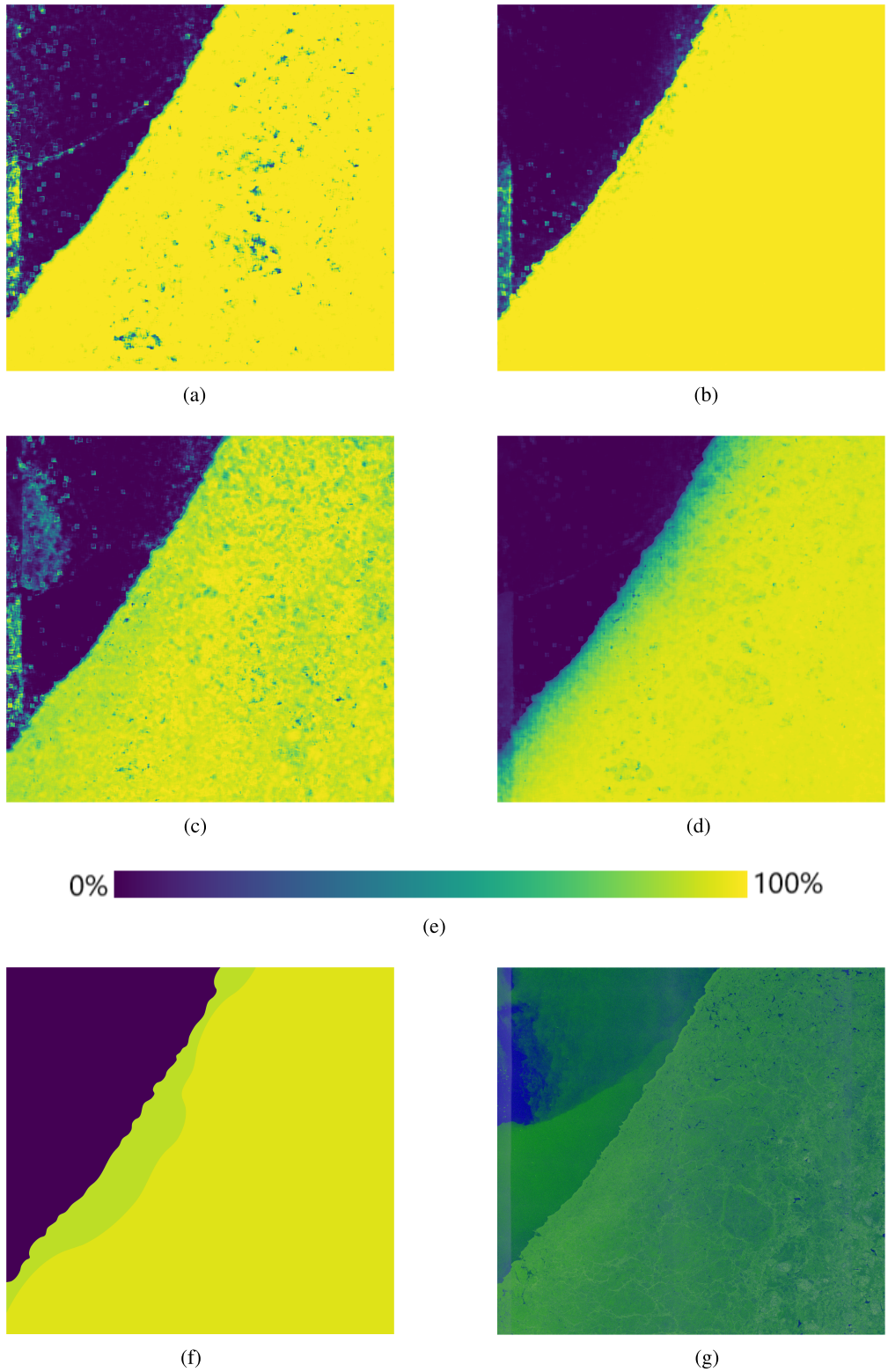

Fig. 5. Zoomed-in view of Sentinel-1 scene from June 17, 2015 (Product Unique ID: D45E) located in the sea between Greenland's west coast and Baffin Island. The size of the depicted area is 171 km × 180 km. (a) CNN model predictions, S1, $t = [T10\%]$. (b) CNN model predictions, S1+AMSR2, $t = [T10\%]$. (c) CNN model predictions, S1, $t =$ [SIC]. (d) CNN model predictions, S1+AMSR2, $t =$ [SIC]. (e) Color bar representing the scale for prediction probabilities in (a)–(d). (f) Ice chart made by DMI ice analysts. (g) RGB composite. Red: HV. Green: HH. Blue: HV/HH.

the "true" probabilities of finding an ice pixel within an ice chart polygon. Under the assumption that most pixels at a scale of the sensor resolution will be either completely ice or open water, the concentration of a polygon corresponds to the probability of a given pixel within that polygon being ice. In addition, we also ensure to reduce the effect of label errors

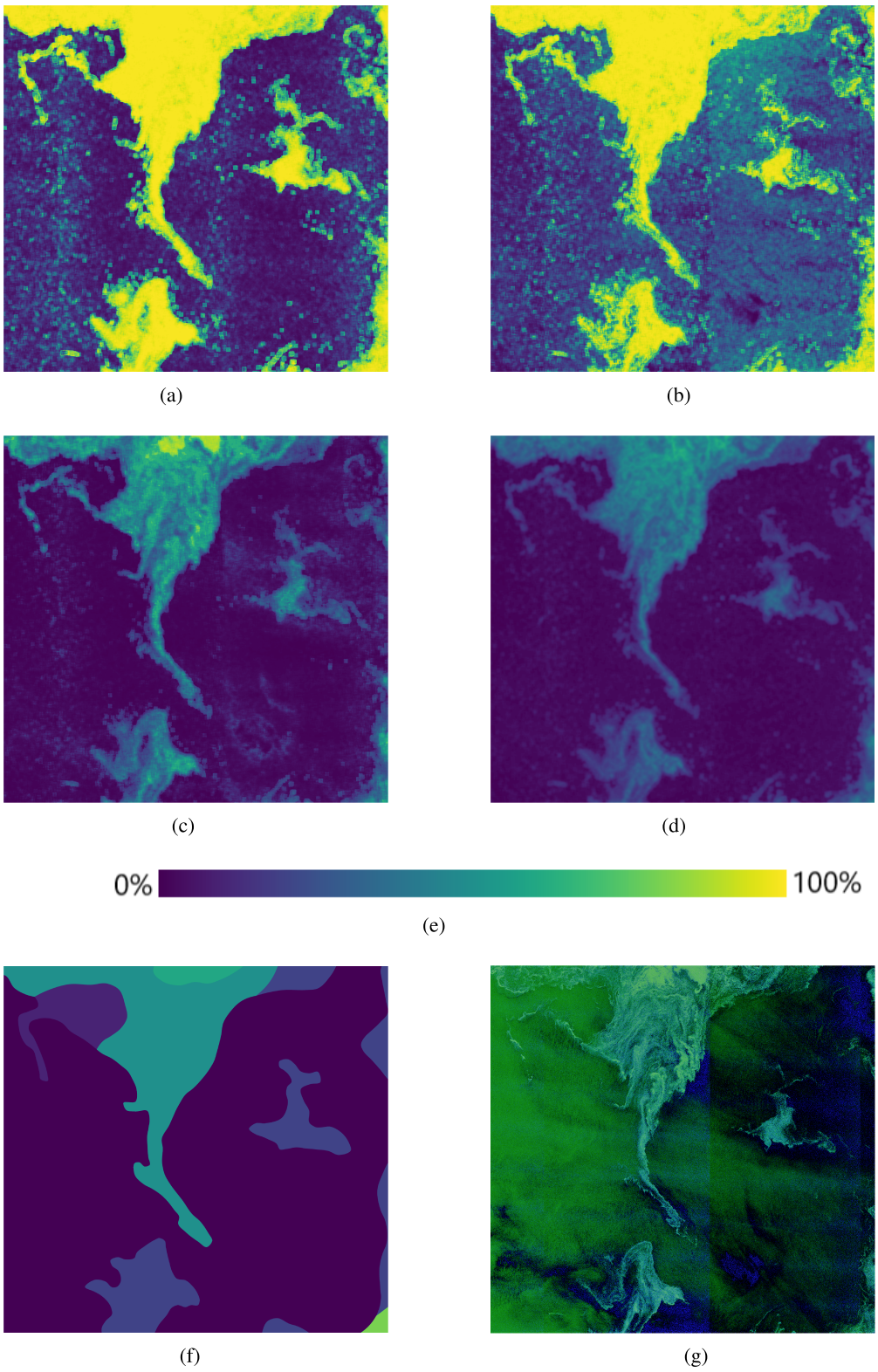

Fig. 6. Sentinel-1 scene used in the above is from the August 22, 2017 (Product Unique ID: F1BD) located in the sea between the east coast of Greenland and the Island Spitsbergen. The figures shows a zoomed-in view of the whole scene of 183 km × 192 km. (a) CNN model predictions, S1, $t = [T10\%]$. (b) CNN model predictions, S1+AMSR2, $t = [T10\%]$. (c) CNN model predictions, S1, t = [SIC]. (d) CNN model predictions, S1+AMSR2, t = [SIC]. (e) Color bar representing the scale for prediction probabilities in (a)–(d). (f) Ice chart made by DMI ice analysts. (g) RGB composite. Red: HV. Green: HH. Blue: HV/HH.

as we "punish" the network less for predicting water pixels in a, e.g., 80% ice zone. The concept of using probabilities as "soft targets" was also used in [32] to distilling the knowledge learned by a large model on a training set into a smaller model, by using the larger models output probabilities. According to [32], the ratio between different class probabilities is said

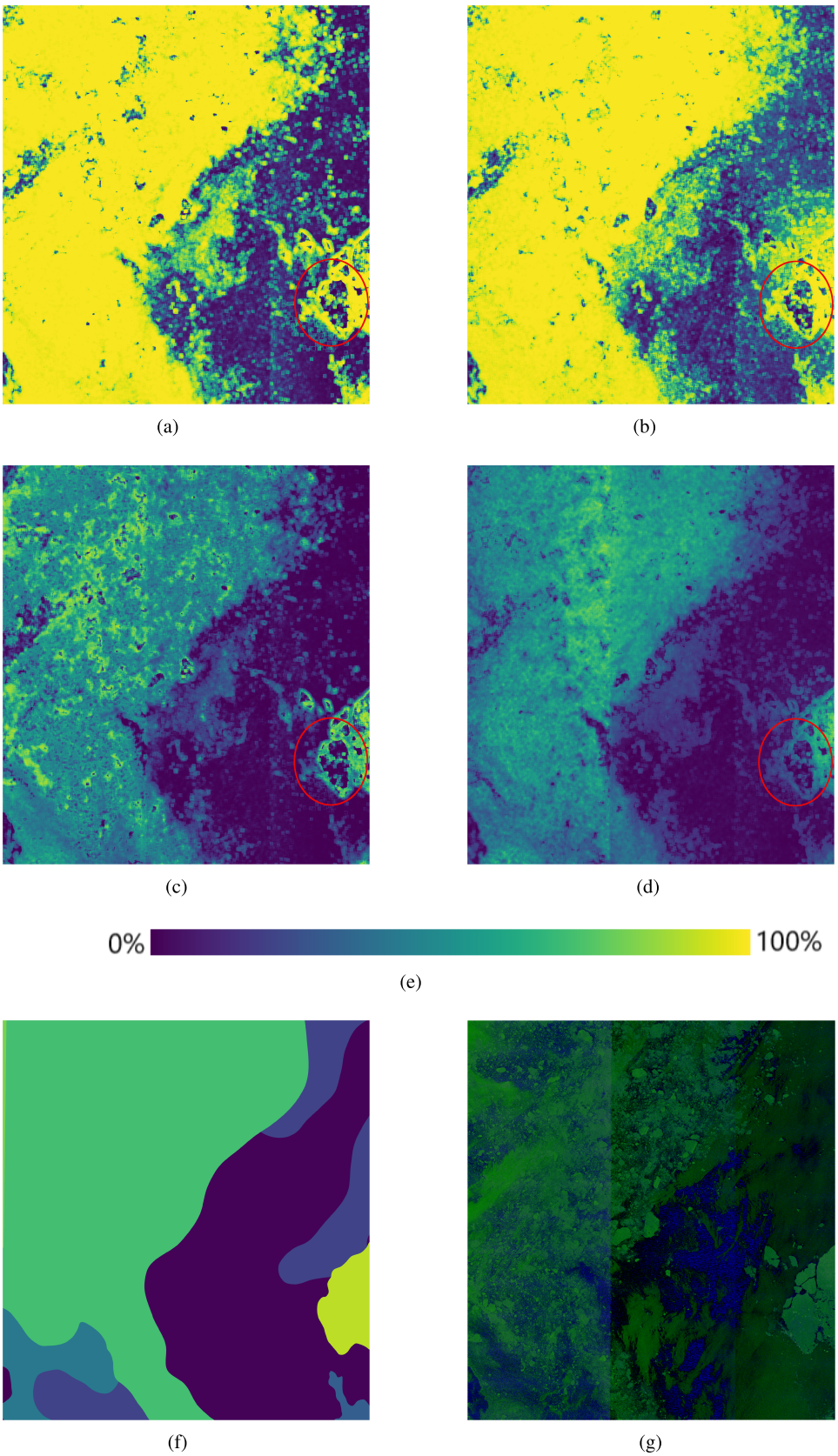

Fig. 7. Sentinel-1 scene used in the above is from August 11, 2017 (Product Unique ID: EF7C) located off the east coast of Greenland near Shannon Island. The figures shows a zoomed-in view of the whole scene of 244 km × 224 km. A red circle in the predictions marks an area where the networks fail to classify ice properly. (a) CNN model predictions, S1, $t = [T10\%]$. (b) CNN model predictions, S1+AMSR2, $t = [T10\%]$. (c) CNN model predictions, S1, $t =$ [SIC]. (d) CNN model predictions, S1+AMSR2, $t =$ [SIC]. (e) Color bar representing the scale for prediction probabilities in (a)–(d). (f) Ice chart made by DMI ice analysts. (g) RGB composite. Red: HV. Green: HH. Blue: HV/HH.

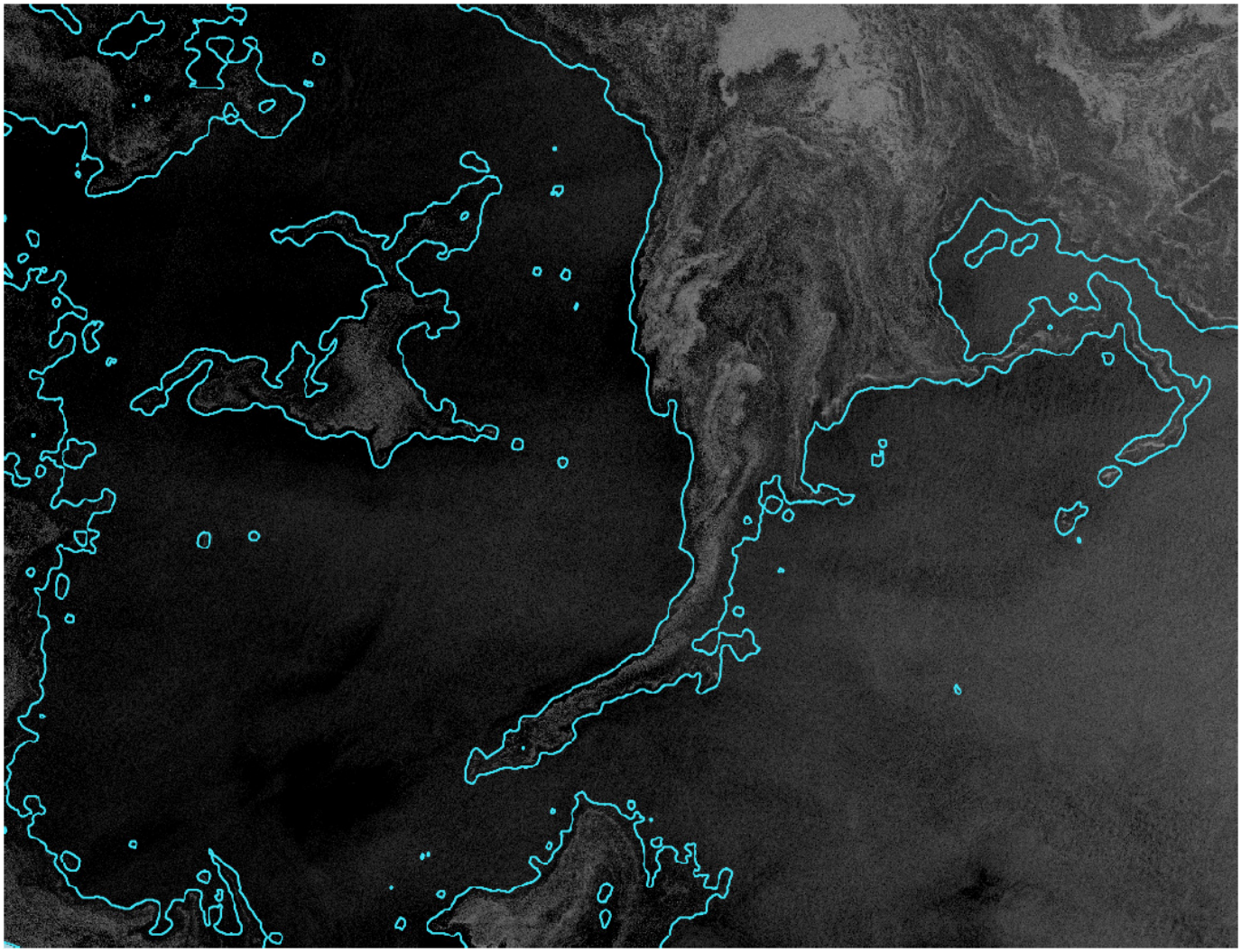

Fig. 8. Polygon outline extracted from CNN predictions (model trained with SICs including both S1 and AMSR2) overlaid on the Sentinel-1 scene from August 22, 2017 (PID: F1BD), HH channel. A 3×3 kernel Gaussian smoothing has been applied to the predictions prior to the extraction of the outline. The image is geographically aligned in QGIS.

to contain richer information about the problem at hand than binary labels. In Section V, we shall investigate the model's ability to align its predictions with the analysts ice charts using either the binary encoding of ice/water or the SIC target probabilities.

V. Results

One might consider different metrics for evaluating models depending on its use for properly addressing the variations in the use of sea ice products. For the majority of users [30], in arctic marine navigation, the main concern is to avoid areas with ice. Another application could be to monitor ice quantities in climate studies or use them as input to forecast models where the main focus would be accurate concentration values. Predicting the ice zone, defined as the boundary between open water and an area with any concentration of ice, is appropriate for the users that wish to avoid sailing in any ice. We, therefore, evaluate the models with: 1) a pixelwise accuracy measure in terms of ice/no-ice zones with ice charts thresholded at 10% alongside and 2) a mean predicted SIC versus analyst estimated SIC as a mean over all polygons within a certain concentration value. Evaluation 1 is shown in Table III as the balanced accuracy, given by the mean of the sensitivity SE and specificity SP, with $\mathrm{SE} = \mathrm{TP}/(\mathrm{TP}+\mathrm{FN})$ and $\mathrm{SP} = \mathrm{TN}/(\mathrm{TN}+\mathrm{FP})$, where TP is the number of true positives, TN is the true negatives, FP is the false positives, and FN is the false negatives. The first subtable provides the performance of two models trained on binary ice masks extracted from thresholding the DMI ice charts at a 10% SIC level. The pixelwise probabilities from the two models in the first subtable have been thresholded at 50% to achieve a binary mask to compare with, since 50% is the middle of the 0/1 labels used for their training. In the second subtable, the two models have been trained to reflect SIC values from the ice charts as the probability (t_i) in (1). These models are trained to provide probabilities that reflect the SIC values. As opposed to a model trained on binary ice labels from thresholding charts at 10% which should have a 50% probability threshold, they have to be thresholded at 10% to give a suitable comparison. By testing at different output probability thresholds for both modeling

approaches, we can confirm that the thresholds are optimal, as seen in Table III. The results in Table III show that we add substantial information to the model when training with SIC values as probabilities rather than binarizing the problem and training with 0/1 labels. Soft targets such as the SIC values here are improving the results of CNN training in line with findings in [32].

The pixelwise accuracy in Table III is showing the performance on predicting whether a pixel belongs to an ice zone or open water. Alternatively, one might be interested in testing the models' overall alignment with analyst assessed SIC values as a performance measure for replicating the analyst's task. This can be done by comparing the mean predictions within each of the test set polygons, as the analyst's estimate also is an average estimation. The polygon average of ice according to the CNN output can be calculated in two ways: either by averaging the probabilities from the CNN or by counting the fraction of pixels assigned a probability higher than 50%. The latter approach has been used for a comparison of the models' alignment with ice chart from analysts in Figs. 3 and 4. Fig. 3 shows the difference between model predictions when we change the target ice mask by thresholding at different SIC values. It is not surprising that thresholding at 50% makes the models' prediction align better with the ice charts than 10%, but in Fig. 4, it is seen that far better results are achieved when training with SIC values as targets in terms of the R^2 measure. Furthermore, it is also advantageous to add AMSR2 data as seen in Fig. 4(b), where higher ice concentrations ($>80\%$) are more accurate. The red points in Figs. 3 and 4 are averages of all polygons in the test set with the size of the dot corresponding to the number of samples N used to calculate the mean. The black line through each point is the standard error calculated as $1.96 * \sigma / np.\text{sqrt}(N)$, where σ is the standard deviation and N is the number of unique ice concentrations in each scene.

Visual examples of the four CNN model variants can be seen in Figs. 5 and 6. It is seen that both inclusion of AMSR2 data and optimizing the CNN with SIC concentrations as cross-entropy probabilities lead to smoother predictions. The noise seen in the predictions in Fig. 5 comes from noise in the first subswath of Sentinel-1's TOPSAR mode. This problem has been reported before and has recently been addressed in [33]. The noise is primarily seen inside the first subswath, near the boundary between first and second subswath, and is occasionally also visible between the other subswaths. In the most recent version of the Sentinel-1 processor (IPF 3.1), the problem has been reduced significantly, but this is only available for data issued after June 6, 2019. The output probabilities in Fig. 5(d) where the model is trained with SIC as a soft probability exhibit visual smoother ice maps that reflect the true ice concentration. Comparing Fig. 5(d) with Fig. 5(b) where binary labels was used, it clearly shows the differences between the two training methods that also are reflected on the plot in Figs. 3 and 4. By measuring accuracy of polygons individually, scenes with poorer performance can be found, such as the one shown in Fig. 7. Other than a similar noise problem as Figs. 6, Fig. 7 also shows a problem (marked by a red circle in the predictions) on a homogeneous ice surface where the center is classified as water. This is likely caused by the ambiguity of the SAR backscatter values on homogeneous ice and open water in windy conditions combined with the CNN not having a sufficiently large field of view to capture the ice edge when classifying pixels in the middle of the ice patch. The CNN is not able to completely suppress this noise in predictions but generally copes better with it when AMSR2 is included and SICs are used as target probabilities [see Fig. 5(d)].

Final operational products can be designed in several ways on the basis of the CNN outputs. One such example is shown on Fig. 8 where the outline of the ice edge (edge between ice in zones and open water). This has been done by applying a 3×3 average filter on thresholded output of the CNN and extracting the vector outline geometry in the QGIS software package.

VI. Discussion and Conclusion

This study presents the first CNN model for combining low-resolution EO data, here AMSR2, with data that are significantly higher in resolution, here Sentinel-1 SAR. The model learns to extract image features from Sentinel-1 SAR imagery and combines it with the low-resolution data from AMSR2 by linear projection similar to a logistic regression. Two modeling schemes are proposed, one where the ice charts are thresholded and used as 0/1 target variables similar to how most segmentation and classification tasks are solved. This showed to be inferior in modeling the ice charts presented here as opposed to the second method where we used the ice concentration values directly in the cross-entropy error function we minimize during training. The binary approach of optimization might still have value in applications where a conservative binary ice mask is sufficient for satisfying user needs though. It can be concluded from the experiments shown here that both the fusion of the two sensors and the cross-entropy optimization on SICs are improving the CNN for prediction of sea ice. Improvements of predictions on high ice concentrations when adding MWR data make sense due to ambiguities in SAR data over homogeneous ice surfaces with open water under various wind conditions.

VII. Future Work

Several limitations may still be addressed in future work. Currently, the noisy first subswath of the Sentinel-1 sensor may be left out under certain conditions, but future approaches could implement promising noise reduction techniques such as in [33] or using data processed with the ESA processor version 3.1 or larger (after June 2019). Furthermore, experiments in the fusion of Sentinel-1 and AMSR2 may be expanded to account for nonlinear relationships by adding additional layers at the end of the current model. An interesting additional study would be the importance of the different AMSR2 brightness temperatures in order to guide future missions in selecting the right frequencies for sea ice applications. There are numerous ways to do this though, and one could choose a leave-one-out or leave-one-in scheme and train CNNs for all possibilities. Another option is first converting MWR brightness temperatures to an estimated ice concentration by a linear projection

such as in [19]. Furthermore, one could calculate features on MWR data, such as channel ratios presented in [20] and [34], to expand the possibilities of feeding the model with MWR derived information. However, it is unlikely to be feasible to do an exhaustive experimental search for an optimal selection of MWR data use.

Data augmentation techniques such as cropping overlapping window samples from the large scenes, combined with larger CNN models, can possibly increase the performance in future experiments. More promising is it though, to incorporate a larger FOV in the network by increasing the dilation rate in the pooling layer of the CNN. Alternatively, one could consider trading a bit of resolution and changing the 40-m pixel spacing to, e.g., 80 m (which is closer to the Sentinel-1 SAR resolution) and thereby immediately double the FOV without increasing computational cost.

Finally, experiments limited to specific regions (e.g., west Greenland) and/or seasons (separating Summer and Winter) could also constitute viable ways for improvements or investigation of the model's performance under different ice conditions.

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ASIP has the aim to partially or fully automate the generation of sea ice information from Copernicus Sentinel-1 SAR imagery. The authors gratefully acknowledge the support of NVIDIA Corporation with the donation of GPUs used for this research, and they acknowledge the Japan Aerospace Exploration Agency (JAXA) for the use of the AMSR2 data.

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